MIL-F-23243A
27 October 1964
SUPERSEDING
MIL-F-23243(SHIPS)
5 July 1962
MIL-M-1250(in part)
27 July 1949

MILITARY SPECIFICATION

FORMING MACHINES, SHEET METAL, ROTARY, MANUALLY OPERATED

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers hand operated rotary, forming machines used in sheet metal shops for beading, crimping, burring, wiring, turning, flanging, double seaming and other related rotary forming operations on sheet metals of the thicknesses (gages) within the rated machine capacities.

1.2 Classification.- The rotary hand operated machines covered by this specification shall be of the following types and shall be suitable for bench mounting (See 6.2):

Type I - Beading.
Type II - Crimping and beading.
Type III - Double seaming.
Type IV - Setting down-machine.
Type V - Combination.
Style A - Regular (throatless).
Style B - Deep throat.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY
MIL-P-18058 - Preparation for Delivery of Machine Tools (Metal and Woodworking), Accessories and Repair Parts.

STANDARDS

MILITARY
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U. S. Military Property.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H-28 - Screw-Thread Standards for Federal Services.

FSC 3441

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(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

3. REQUIREMENTS

3.1 Design.- The rotary forming machines covered by this specification shall be new, the manufacturer's latest models, which have proved satisfactory in operation and dependable in service.

3.2 Construction.- Construction shall be free from any characteristics or defects which may render the machine unsuitable for the purpose intended. Machine shall be of rugged construction. All materials used shall be as herein specified and those not specified shall be entirely suitable for the purpose intended. Machine shall be complete so that it can be used for any operation for which designed.

3.2.1 Casting and forgings.- All castings and forgings shall be free from defects, scale and mismatching. In no event shall such processes as welding, peening, plugging or filling with cold solders or metallic paste be used on castings or forgings for reclaiming any part of the machine without authorization.

3.2.2 Welding, brazing or soldering.- Welding, brazing or soldering shall be employed only where those operations are included in fabrication of the original design. These operations shall not be employed as repair measures for defective parts.

3.2.3 Fastening devices.- All screws, pins, bolts and similar parts shall be installed with means for preventing loss of tightness. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.2.4 Threads.- All threaded parts shall be in the inch system and shall conform to H-28.

3.2.5 Lubrication.- Bearings, revolving shafts and moving parts shall be provided with ample and suitable means of lubrication, as required.

3.2.6 Interchangeability.- All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or machine.

3.2.7 Surfaces.- All surfaces of all castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free from sand, dirt, fins, sprues, scale, flux or other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled.

3.2.8 Painting.- Unless otherwise specified (see 6.2), the machine shall be painted in accordance with the manufacturer's standard commercial practice.

3.3 Detailed requirements.-

3.3.1 Type I.- Type I machines to be supplied under this specification shall be the hand-operated type. Machine shall be capable of forming beads on cylindrical or flat pieces of sheet metal of the rated capacity thickness. Type I machines shall be similar to figure 1 and shall conform to the requirements of table I for the capacity specified (see 6.2). The machine shall consist of a mounting base, shafts, gears, rolls and the necessary operating handles, gages, guards, and adjusting mechanism required.

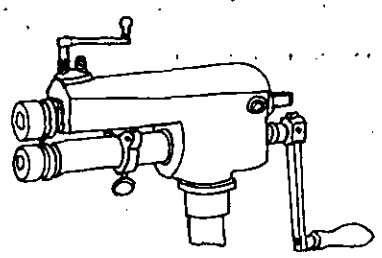
3.3.1.1 Base.- Each type I machine shall be provided with a stand or standard for bench mounting. Stand or standards shall be designed for mounting the machine and providing bench clearance for all intended operations. Stands or standards shall be cast-iron, semi-steel or fabricated steel weldments.

3.3.1.2 Shafts.- Shafts shall be of steel machine finished. Shafts shall be driven by hand crank and gears and shall be fitted to bearings.

3.3.1.2.1 Bearings.- Bearings shall be made of machined bronze, cast-iron, or other bearings and shall be fitted to the shafts to insure smooth operation and long life. The lower shaft bearings shall be carried in the horn.

3.3.1.3 Gears.- All gears shall be cast-iron or machine cut steel and shall be totally enclosed or guarded to protect the operator.

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SH 7042

Figure 1 - Type I forming machine, beading.

Table I - Type I forming machine, beading.

Capacity mild steel (gage) inch	(24) 0.0239[1/]	(22) .0299[1/]	(20) .0359[1/]	(18) .0478[1/]
Distance between shaft centers (min) inches	1-7/8	2-1/3	4-1/8	2-2/3
Distance center of rolls to gage (min) inches	6-1/4	7-1/3	29-1/4	12-3/4
Shipping weight, pounds (within plus or minus 10 percent.	55	94	400	185

[1/] Manufacturer's standard gage.

3.3.1.4 Rolls.- Rolls shall be made of good quality steel machine finished and hardened to 52 to 62 Rockwell "C." Rolls shall be of the type herein specified or as specified (see 6.2). Rolls furnished with this type machine shall satisfactorily perform the operations for which designed and shall comply with the applicable test performances specified in 4.2.2.

3.3.1.5 Adjustment.- The upper roll shall be adjustable by means of a quick acting crank screw for controlling the roll adjustment.

3.3.1.6 Gage.- Machine shall be provided with an adjustable gage to govern the position of the bead or other operations on the work piece. Gage shall be smooth finished and fitted to machine.

3.3.1.7 Equipment and accessories.- Type I machines shall be furnished with the following beading rolls for the capacity specified (see 6.2):

For 18, 20 and 22 gage machines

(a) One pair of 3/8-inch single beading rolls.
(b) One pair of 3/4-inch or 1-inch O.G. beading rolls.

For 24 gage machines

(a) One pair of 1/4-inch single beading rolls.
(b) One pair of 3/4-inch O.G. beading rolls.

3.3.1.8 When specified, (see 6.2), the following accessories shall be furnished:

(a) One pair of crimping rolls with gage.
(b) One pair of wiring rolls.
(c) One pair of edging rolls.
(d) One pair of turning rolls.

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3.3.2 Type II.- Type II machines shall be suitable for crimping and beading operations of sheet metal cylinders of a rated capacity thickness and shall also be capable of performing both operations simultaneously. Type II machines shall be similar to figure 2 and shall conform to the requirements of table II, for the capacity specified (see 6.2) and shall comply with the test requirements of Section 4. Type II machines shall consist of a frame supporting two parallel shafts, gears, stand, operating crank, adjusting device and gages.

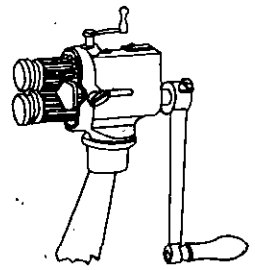


Figure 2 - Type II forming machine, crimping and beading.

Table II - Type II forming machine, crimping and beading.

Capacity in mild steel (gage) inch	(24) 0.0239[1/]	(20) .0359[1/]	(18) .0478[1/]
Distance between shaft centers inches (min)	1-1/2	1-3/4	2-5/8
Gearing ratio	1 - 1	3 - 1	4 - 1
Shipping weight, pounds (within plus or minus 10 percent)	40	50	140

1/ Manufacturer's standard gage.

3.3.2.1 Frames.- Frames shall be a rugged casting of such design to support the shafts, drive mechanism and provide a suitable base connection for bench mounting.

3.3.2.2 Shafts.- Shafts shall conform to the requirements of 3.3.1.2.

3.3.2.3 Bearings.- Bearings shall be of machined bronze, cast iron or other bearing of rugged design and shall be fitted to the shafts to insure smooth operation and long life.

3.3.2.4 Rolls.- Rolls shall be made of a good grade steel machine finished, and hardened to 52 to 62 Rockwell "C".

3.3.2.5 Adjustment.- The upper roll shaft shall be provided with a quick acting crank screw or equal means of adjustment. This adjusting mechanism shall be designed to provide the necessary pressure to the rolls to form the required shapes.

3.3.2.6 Gage.- The machine shall be provided with a gage mounted to the frame in such a manner as to govern the length of crimp being formed on the work piece.

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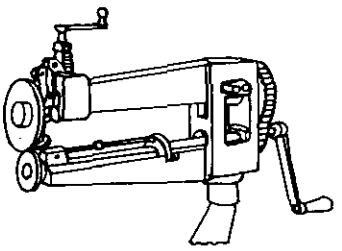
3.3.2.7 Gears.- All gears used in the assembly of type II machines shall be machine cut steel or a good grade of cast iron.

3.3.2.8 Stands.- Type II machines shall be provided with a bench type stand, of the screw clamp type. Stand shall be shaped to bring the machine forward to provide ease of machine operation.

3.3.2.9 Equipment and accessories.- Each type II machine shall be furnished complete with the following equipment. Extra rolls shall be as specified (see 6.2).

(a) One pair crimping rolls (straight).
(b) One pair beading rolls.
(c) One pair spacing collars.
(d) One stand for bench mounting.

3.3.3 Type III.- Type III machines shall be suitable for finish seaming flat bottoms on straight or flared sheet metal cylinders after the edges have been flared and set down. Type III machines shall consist essentially of a frame, two parallel shafts, gears, adjusting device, rolls and stand. Type III machines shall be similar to figure 3 and shall conform to the requirements of table III for the capacity specified (see 6.2).



SH 7043

Figure 3 - Type III forming machine, double seaming.

Table III - Type III forming machine, double seaming.

Capacity in mild steel (gage inch)	(26) 0.0179[1/]	(24) .0239[1/]	(20) .0359[1/]
Throat depth min, inches	15	15	16
Diameter of lower seaming roll max, inches	2-3/4	2-3/4	6
Shipping weight, pounds (within plus or minus 10 percent)	110	155	500

[1/] Manufacturer's standard gage.

3.3.3.1 Frame.- Frame shall be of substantial design and shall rigidly support the two parallel shafts.

3.3.3.2 Shafts.- Shafts shall be of high grade steel, machine finished. Shafts shall be driven by hand crank and gears and shall be fitted to bearings, to provide smooth and easy operation at rated capacity.

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3.3.3.3 Bearings.- Bearings shall be of machined bronze or cast iron and shall be fitted to the shafts.

3.3.3.4 Gears.- Gears shall be of cast iron or machine cut steel and shall be totally enclosed, and guarded to protect the operator.

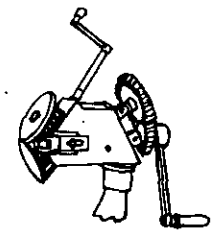
3.3.3.5 Adjustment.- The upper roll shall be adjustable by a quick acting crank screw, for controlling the roll adjustment. Fast adjustment of roll for first and second seaming operations with positive stops for correct alignment shall be provided.

3.3.3.6 Rolls.- Upper rolls shall be made of steel machine finished and hardened to 52 to 62 Rockwell "C." Lower seaming roll capacity shall conform to table III for the capacity of type III machine specified (see 6.2).

3.3.3.7 Stands.- Machines equipped with a socket plug type base and shall be furnished with a bench type pedestal or standard.

3.3.3.8 Equipment and accessories.- Each type III machine shall be furnished complete with rolls and stand or base.

3.3.4 Type IV.- Type IV machines shall be capable of squeezing or pinching together the flanges of the bottoms and sides of cylindrical containers or like items prior to double seaming operations. The type IV setting down machines shall consist of a frame, shafts, rolls, hand crank, gears and stand. Type IV machines shall be similar to figure 4, and shall comply with the requirements of table IV for the capacity of machine specified (see 6.2).



SH 7044

Figure 4 - Type IV forming machine, setting down.

Table IV - Type IV forming machine, setting down.

Capacity in mild steel, (gage) inch	(24) 0.0239[1/]	(20)..0359[1/]	(18) .0478[1/]
Maximum width edge will close at least inches	3/16	3/8	1/2
Minimum width edge will close inches or less	1/8	3/16	1/4
Shipping weight, pounds (within plus or minus 10 percent)	50	96	96

[1/] Manufacturer's standard gage.

3.3.4.1 Frame.- The frame shall be of substantial design, entirely suitable for the purpose intended. It shall support the rolls, shafts, gearing and mounting base.

3.3.4.2 Shafts.- Shafts shall be of steel, machined and finished. Shafts shall be driven by hand crank and gears. Shafts shall be fitted to bearings to provide smooth and easy operation at rated capacity.

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3.3.4.3 Bearings.- Bearings to support the shafts and rolls shall be machined bronze or cast iron, and shall be fitted to the shafts.

3.3.4.4 Gears.- Gears shall be cast iron or machine cut steel or other suitable metals which have proved entirely satisfactory in operation and dependable in service.

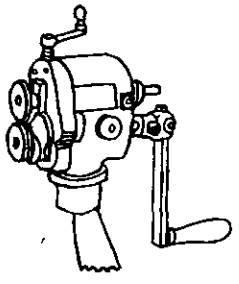
3.3.4 5 Rolls.- Rolls shall be made of steel machine finished and hardened to 52 to 62 Rockwell "C." Roll faces shall be finished on an angle and shall be mounted inclined inwardly on the same angle as on the finished faces so that the faces of the two rolls shall have full parallel bearing to perform the setting down operations required.

3.3.4.6 Adjustment.- Adjustment of rolls shall be provided by a quick acting crank screw, for controlling the roll adjustment.

3.3.4.7 Stands.- Unless otherwise specified (see 6.2), type IV machines shall be provided with a base for bench mounting.

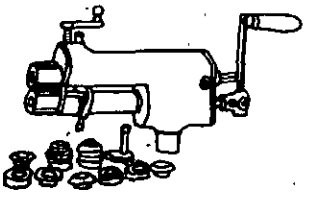
3.3.4.8 Equipment.- Type IV machines shall be furnished complete with rolls and base.

3.3.5 Type V.- Type V machines shall be capable of performing turning, burring, wiring, edging, flattening or flanging operations on sheet metal of the rated capacity thickness when used with the applicable forming rolls. Type V machines shall be similar to figure 5 and shall conform to the requirements of table V for the style and capacity specified. Machines of 24 capacity gage and 22 capacity gage shall be furnished in either style A or style B, as specified (see 6.2). Type V machines shall consist essentially of a frame, shafts, gearing, adjusting device, gages, rolls and stands.



SH 7045

Style A regular (throatness).



SH 7046

Style B (deep throat).

Figure 5 - Type V forming machine, combination.

3.3.5.1 Frames.- Frames shall be castings of substantial design to support the shafts, bearings, gears and to protect the working parts from dirt and dust.

3.3.5.2 Shafts.- Shafts shall be made of steel machined and finished. Shafts shall be driven by hand crank and gearing and shall be fitted to suitable bearings.

3.3.5.3 Bearings.- Bearings shall be fitted to the shafts and shall be of bronze, cast iron or other metals which will insure long life and smooth operation.

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Table V - Type V forming machine, combination.

	Style				
	A Regular (throatless)	B (deep throat)	A Regular (throatless)	B (deep throat)	A Regular (throatless)
Capacity in mild steel, (gage) inch	(24) 0.0239 [1/]	(24) .0239 [1/]	(22) .0299 [1/]	(22) .0299 [1/]	(18) .0478 [1/]
Distance between shaft centers inches (min)	2	1-7/8	2	1 - 1	2-3/4
Ratio of gearing	1 - 1	1 - 1	1 - 1	1 - 1	2 - 1
Maximum wire diameters that can be used at least (inches)	3/16	3/16	5/16	1/4	5/16
Depth of throat (center of rolls to gage) (min.) inches	----	6-1/4	----	6-1/4	----
Shipping weight, pounds, within plus or minus 10 percent	30	55	35	55	105

[1/] Manufacturer's standard gage.

3.3.5.4 Gears.- All gears shall be cast iron or machine cut steel and shall be totally enclosed or provided with a guard to protect the operator.

3.3.5.5 Adjustment.- Adjustment of the rolls shall be provided by a crank screw or similar means. Additional adjustments to provide vertical alignment of rolls for proper clearance shall be provided.

3.3.5.6 Rolls.- Rolls shall be made of steel, finished and hardened to 52 to 62 Rockwell "C." The rolls furnished with each type V machine shall be of such design as will satisfactorily produce the type of work for which designed. Each type V machine shall be furnished complete with the following rolls and gages as standard equipment:

(a) One pair of turning rolls (thin).
(b) One pair of burring rolls.
(c) One pair of wiring rolls.
(d) Gages for use with above rolls.

3.3.5.6.1 *Additional quantities of rolls shall be as specified* (see 6.2).

3.3.5.7 Stands.- Unless otherwise specified (see 6.2), each type V machine shall be provided with a stand or base for bench mounting.

3.3.5.8 Equipment.- Each machine shall be furnished complete with base stand and the rolls listed in 3.3.5.6 and such additional accessories and wrenches as are normally furnished commercially with this type machine.

3.3.5.9 Additional equipment.- When required (see 6.2), the following equipment shall be furnished as specified:

(a) One pair of flanging rolls (for style A machine).
(b) One pair of elbow edging rolls (for style A machine).
(c) One pair of flattening rolls (for style A machine).
(d) One circle flanging attachment for burring discs 5 to 24 inches (18 gage only) (for style A machine).
(e) One pair of O.G. beading rolls (for style B machine).
(f) One pair of single beading rolls (for style B machine).
(g) One pair of cutting rolls (for style B machine).

3.3.5.9.1 All rolls shall be furnished complete with adaptable gages.

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3.4 Nameplate.- Unless otherwise specified (see 6.2), a corrosion resistant metal nameplate, permanently and legibly marked with the following information in accordance with MIL-STD-130 shall be securely attached to each machine:

Manufacturer's Name.
Manufacturer's Model Number.
Manufacturer's Serial Number.
Contract Number.
Date of Manufacture.
US.

3.5 Identification plate.- When specified (see 6.2), all machines supplied shall carry a separate identification plate of corrosion resistant metal permanently and legibly marked with the following information. Figures shall be at least 3/16 of an inch in height.

Production Equipment Code Number (to be supplied in contract)
Activity Identification Number (to be inscribed by receiving activity)

3.6 Handbook of instructions.- Unless otherwise specified (see 6.2), the contractor shall furnish two copies of manufacturer's standard handbook.

3.7 Workmanship.- Workmanship of the rotary forming machines and accessories shall be of the quality prevailing among manufacturers normally producing rotary forming machines of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2.1 Examination.- All machines shall be examined to determine compliance with the requirements of this specification. Any deviation from these requirements shall be cause for rejection.

4.2.2 Tests.- Each machine shall be operated at maximum rated capacity in each of operations for which designed and shall produce a satisfactory design shape on the sheet metal being formed. Machines shall operate smoothly at rated capacity.

4.2.3 Examination of preservation, packaging, packing and marking for shipment and storage.- Preservation, packaging, packing and marking for shipment and storage shall be examined to determine conformance with the documents referenced in Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing.- Unless otherwise specified (see 6.2), preservation, packaging, and packing shall be in accordance with MIL-P-18058. Level of preservation and packaging, and level of packing shall be as specified (see 6.2).

5.2 Marking.- Unless otherwise specified (see 6.2), marking shall conform to MIL-STD-129.

6. NOTES

6.1 Intended use.- The rotary forming machines furnished under this specification are designed for hand operation in the fabrication of cylindrical containers and covers, air ducts, and like cylindrical items of sheet.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type and style required (see 1.2).

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(c) Capacity required (see applicable table) (tables I, II, III, IV, V).
(d) Special or extra rolls required (see 3.3.1.4, 3.3.1.8, 3.3.2.9, 3.3.5.9).
(e) Extra equipment or accessories (see 3.3.5.9).
(f) Forward and reverse rotation and gaging from each side, if required (see 3.3.5).
(g) Painting, if different (see 3.2.8).
(h) Nameplate, if different (see 3.4).
(i) Identification plate, if required (see 3.5).
(j) Handbook requirement, if different (see 3.6).
(k) Preservation, packaging and packing, if different (see 5.1).
(l) Level of preservation and packaging and level of packing (see 5.1).
(m) Marking, if different (see 5.2).

6.3 Superseding data. This specification supersedes and replaces the following kinds and types of machines of MIL-M-1250.

MIL-F-23243			MIL-M-1250	
Type	Style	Machine	Kind	Type
I	---	Beading	A	I
V	B	Burring	C	---
II	---	Crimping	D	---
III	---	Double seaming	E	---
V	A	Flanging	G	---
IV	---	Setting down	M	---
V	A	Turning	O	---
V	A	Wiring	P	---
V	B	Combination	Q	I

Custodians:
Army - WC
Navy - SH
Air Force - 84

Review activities:
Army - WC
Navy - SH, YD
Air Force - 84

User activities:
Army - MO
Navy - WP

Preparing activity:
Navy - SH
(Project 3441-0006)

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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